

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in linkedin.com/in/muhammad-haider-zaidi
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python (Fluent, large codebase), C++, SQL, Javascript
Deep Learning & AI: PyTorch, TensorFlow, Keras, YOLO, CNNs, Transformers, LLMs, Numerical Algorithms
Computer Vision & Image Proc.: Object Detection, Segmentation, Real-time Systems, Image Analysis
MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, TensorRT, MLOps Pipelines, AI-assisted development
Architecture: Client/Server, Multi-threaded, Hardware Interfacing (Cameras, Sensors), Communication Protocols
Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing, Data Visualization

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for high-precision metrological applications, improving model performance and training efficiency for enhanced accuracy.
- Explored novel AI architectures and physics-informed hybrid loss functions, developing optimized algorithms for improved model accuracy and stability across various layers of the software stack.
- Developed dashboards for real-time data visualization and continuous model performance monitoring, contributing to software quality and optimization.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized vitiligo segmentation accuracy to 94% and increased inference speed by 5x using advanced AI algorithms, deployed within a client/server architecture with Docker and FastAPI.
- Engineered a real-time tennis foul detection system with near-hardware programming (TensorRT), reducing compute cost and processing time by 6x for enhanced performance and scalability.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure for scalable and reliable AI deployments, reflecting an AI-native engineering mindset.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed full-stack web applications providing AI-based services using Python (Flask), adhering to clean code principles and a layered architecture.
- Designed and implemented an AI chatbot application, leveraging modern LLM frameworks and APIs to integrate advanced AI functionalities and communication protocols.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Developed and automated ETL pipelines for ingesting and processing structured and unstructured data, ensuring high data quality for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble, integrating advanced deep learning models and custom feature engineering (AlphaFold) to optimize protein function prediction.
- Increased model accuracy by 4% through custom feature engineering and algorithmic design, demonstrating continuous optimization of functionalities.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed and implemented a real-time Bitcoin price forecasting system using advanced time-series analysis and machine learning models, optimizing for prediction accuracy.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed a real-time signal processing system, integrating temporal dependencies for AI-driven music generation, demonstrating interaction with specific hardware components.
- Implemented transformer-based models for sequence prediction in audio generation, leveraging modern ML/AI frameworks for complex algorithmic design.

Hugging Face

Open-Source

Contributor

- Co-authored technical documentation and implementation for Zero-shot Image Classification on the Hugging Face Hub, showcasing expertise in computer vision and ML/AI frameworks.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling, focusing on numerical algorithms and software optimization.
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems, showcasing hands-on experience with modern ML/AI frameworks.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**
Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis and algorithm design.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python